Prodigy Anesthesia Quick Review

Ventilator

Ventilators are cycled from inspiration to expiration or from expiration to inspiration through one of what four variables?	1. Volume 2. Pressure 3. Time 4. Flow *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 55.*
What is the most common mode of ventilation?	Volume controlled ventilation *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 55.*
Describe volume controlled ventilation.	In volume controlled ventilation, the ventilator controls the inspiratory flow and delivers a preset tidal volume with each breath. Tidal volume and inspiratory flow are constant, however the airway pressure required to deliver the preset volume varies depending on the patient's resistance and compliance. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 57.*
Does peak inspiratory pressure remain constant during volume controlled ventilation?	No. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 57.*
Describe pressure controlled ventilation.	With pressure controlled ventilation, there is a target gas delivery pressure established. Tidal volume varies from breath to breath depending upon the patient's resistance and compliance. At the beginning of inspiration, the ventilator rapidly increases the pressure to the determined level and maintains that pressure until the beginning of exhalation. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 56.* *Dorsch JA, Dorsch SE. Understanding Anesthesia Equipment. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2008: 317.*
During pressure controlled ventilation, if airway resistance increases, or compliance decreases, what happens to the patient's tidal volume?	The tidal volume decreases. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 174.*

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How does PEEP (Positive End Expiratory Pressure) improve oxygenation?	1. Decreases the physiological shunt 2. Prevents alveoli from collapsing 3. Restores the functional residual capacity. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 57.*
What is plateau pressure?	Plateau pressure is a measurement of lung compliance during an inspiratory hold. By holding inspiration for a moment, gas flow drops to zero and airway pressure is allowed to reach a contant value. This value, known as plateau pressure represents the elastic recoil of the lungs at end-inspiration. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 446-447.*
What is static compliance? How is static compliance calculated?	Static compliance represent lung compliance during a period of zero gas flow, such as during an inspiratory hold. Static compliance = Tidal Volume/(plateau pressure - PEEP). The normal static compliance in adults is between 35 and 100 mL/cm H2O. In children, it is normally 15 mL/cm H2O or higher. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 446-447.*
When is inspiratory gas flow the highest during pressure controlled ventilation?	At the beginning of inspiration. Inspiratory flow accelerates rapidly at the beginning of inspiration to attempt to reach the target set pressure. Once this pressure is achieved, flow rapidly decreases. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 57.*
Where on the anesthesia machine is a detachable PEEP valve placed?	PEEP valves are placed on the expiratory limb of the breathing system. Most modern ventilators have built-in PEEP valves, however. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 58.*
What is the result of using a PEEP valve on a spontaneously breathing patient?	There is an increased work of breathing *Dorsch JA, Dorsch SE. Understanding Anesthesia Equipment. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2008: 201.*
Are PEEP valves unidirectional or bidirectional flow valves?	They may be unidirectional or bidirectional. PEEP is only produced when gas flows in the correct direction through the valve. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 59.*

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If a unidirectional PEEP valve is incorrectly placed on the inspiratory limb opposing gas flow, what is the result?	An obstruction occurs resulting in the patient not being ventilated. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 58.*
If a unidirectional PEEP valve is incorrectly placed on the expiratory limb opposing gas flow, what is the result?	Severe barotrauma can occur *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 58.*
If a bidirectional PEEP valve is misplaced on either limb of the system opposing gas flow, what is the result?	There will be no obstruction, but no PEEP will be delivered to the patient. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 58.*

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